

Exact flavor tensors, coupled running and mechanism selection

1,891 exact quartic products, finite-mass hierarchy exclusion and a CP-quality interaction

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A specified 92-parameter one-loop system, canonical scalar matching, finite-mass kinetic response and resolved vacuum comparisons. Exact certificates and numerical benchmark conclusions retain separate scopes.

Central results and declared scope

The complete minimal nonet action and its Gaussian finite-mediator extension now have exact scalar-loop proofs and a coupled one-loop evolution system. Every product of the 52 source quartics is proved coefficient by coefficient; adjoining all four real Higgs components gives a 61-quartic algebra with 1,891 proved products. The specified renormalizable soft-current class has 92 running nonconstant parameters. Its evolution includes gauge couplings, all eight allowed Yukawa coefficients, two universal heavy-quark masses, source masses, Higgs portals and nine mediator parameters. A complete finite census of coordinate-supported adjoint-pair subalgebras, exact current anomalous coefficients, leading heavy-scalar matching with canonical normalization, and finite quark-metric response accompany the evolution.

Three further results select mechanisms rather than nominate another angle. A pure traceless Hermitian adjoint with universal bare masses obeys the exact light-mass bound $m_2/m_1 < 1 + m_2/m_3 < 2$. A family-invariant CP-odd quartic gives an allowed dimension-eight CP-even Yukawa interaction that shifts the strong-CP mass phase. Finally, an explicitly calculated, locally stable CP-conserving stationary point lies below the retained CP-breaking branch of the complete minimal action. These are different statements: the mass bound is exact, the operator covariance is exact, and the stationary-point energy comparison is numerical with a resolved gap.

The target angle and golden CKM coefficient are not used in these new calculations. Neither is predicted. The electromagnetic gauge boundary remains free. Complete two-loop fermion matching, a complete higher-operator classification, a phenomenological threshold-matched CKM flow and a proof of the absolute global vacuum are not outputs of this chapter. Each approximation below is part of the result's definition.

Exact tensor proof of the full scalar algebra

The earlier table obtained by numerical rational reconstruction is now independently certified. Use an unnormalized Hermitian adjoint basis with real off-diagonal entries equal to one, imaginary off-diagonal entries proportional to $i\sqrt{3}$, and diagonals (1,-1,0) and (1,1,-2). The kinetic metric is $G = \text{diag}(2,6,2,6,2,6,2,6)$. In this basis all invariant projector tensors lie in $Q(\sqrt{5})$. Numerical projectors only propose integer coefficient pairs (a,b)/80 representing $(a + b\sqrt{5})/80$; the subsequent proof does not use a reconstruction tolerance.

Exact arbitrary-precision integer identities establish covariance under the four defining family generators and the specified generalized CP operation, metric idempotence, pairwise orthogonality, completeness, and ranks 1,8,8,9,10. This fixes the CP-even 5 plus 5-prime sum rather than an arbitrary multiplicity-space matrix. The source polynomial tensors have common denominator 240. Their kinetic Hessian contractions are computed symbolically. For every symmetric pair i,j, the complete polynomial identity $\text{Tr}(G^{-1} \text{Hess } P_i G^{-1} \text{Hess } P_j)/2 = \sum_k A_{kij} P_k$ is checked, including all monomials generated outside the proposed basis support.

The 52-source table passes all 1,378 products and 19,227 nonzero monomial comparisons with zero exact residual. Adding $H^\dagger H$ squared and the eight source-quadratic Higgs portals gives 61 quartics, 1,891 products and 20,538 nonzero monomial comparisons. A finite-field evaluation matrix supplies an exact independence witness; tests also verify rank 61 for the extended basis. The finite-field argument proves characteristic-zero independence because the nonzero minor is evaluated at a prime with valid denominators and a specified image of $\sqrt{5}$. It does not replace the coefficient identities.

The proof lives in `python/perfectpower/exact_nonet_algebra.py`.
`receipts/m22_interactions/nonet_exact_tensor_proof.json` stores the exact projector pairs and the source rank witness; `nonet_higgs_exact_loop_algebra.json` stores the extended rational multiplication table. These are exact Python certificates, not Lean kernel proofs. In physical normalization, $16 \pi^2 \beta(V4) = \text{Tr}[(G\text{-inverse Hess } V4)^2]/2$. The familiar one-field coefficient 72 for x to the fourth and eight-component coefficient 128 for the squared adjoint norm provide normalization checks.

Complete restricted mechanism census and current running

Consider the nine-dimensional adjoint-pair quartic space consisting of each sector's radial square and finite 8-prime self invariant, the product of the two radial norms, the ordinary adjoint overlap, and the three finite cross channels. Every one of its 512 coordinate-support subsets is tested against the exact multiplication table. Exactly 14 are closed, including the empty subspace. Among these subsets, any one containing an orientation-sensitive ordinary or finite cross channel must contain all nine operators. This is a complete coordinate-support classification within the stated ambient algebra. It does not classify arbitrary rotated subspaces or nonlinear invariant coupling varieties.

Exact generated-subalgebra calculations separately retain important rotated relations. Radial seeds close in dimension two. A single finite cross channel together with the two radial seeds generates dimension nine, as does the ordinary cross channel. Equal coefficients on all three finite channels implement the continuous $SU(3)$ 27-plet relation and generate a five-dimensional algebra. That algebra survives scalar renormalization, but the previously calculated fixed-spectrum orientation obstruction still prevents it from isolating the desired interior CP frame. Closure alone does not imply predictive mixing.

For the Gaussian UV completion, write the dimension-one currents as $g_{uR} S_R q_{uR} + g_{dR} S_R q_{dR}$ and the mediator mass as M_R^2 . The exact finite-8 self invariant acts on the three finite current irreps with eigenvalues $k_R = (248/15, 16/5, -112/15)$, ordered 8-prime, 9, and the CP-paired five-dimensional irreps. The identities are proved as exact field matrices, not inferred from sample vectors. With radial coefficients λ_f , finite-self coefficients κ_f , cross coefficients λ_R and adjoint Yukawa coefficients a_f , the one-loop coefficients are as follows, with a common factor $16 \pi^2$ suppressed.

$\beta(g_{uR}) = (16 \lambda_u + k_R \kappa_u + 12 a_u^2) g_{uR} + 4 \lambda_R g_{dR}$.
 Interchange u and d for $\beta(g_{dR})$. $\beta(M_R^2) = 4(g_{uR}^2 + g_{dR}^2)$.
 Differentiating the tree exchange coefficient $-g_{uR} g_{dR}/M_R^2$ then gives its running, including the mass derivative. Generic finite-self anisotropy splits equal exchange coefficients because the three k_R differ. Vanishing anisotropy with unified cross coefficients is a genuine restricted exception. No conclusion excludes all possible tuned or nonlinear relations.

An exact finite-mass exclusion for a pure adjoint source

The simplest source-label reduction removes the two singlets from each nonet. Its complete minimal scalar action has 15 coefficients, including 12 quartics and the Higgs; the corresponding nonconstant running counts are 24 without, or 33 with, the Gaussian finite mediators. This simplification cannot supply a strong hierarchy under universal canonical fermion matching.

Let D have blocks $(a I_0; C, M I)$, with a, M positive and C traceless Hermitian. For nondegenerate light singular values $0 < m_1 < m_2 < m_3 < \min(a, M)$, each squared source eigenvalue is $f(m) =$

$(a^2 - m^2)(M^2 - m^2)/m^2$. The largest source magnitude belongs to m_1 . Tracelessness in three dimensions forces that magnitude to equal the sum of the other two magnitudes, including either possible overall sign. The following two exact positive differences establish the bound without a large- M approximation.

$f(m_1) - (m_2/m_1)^2 f(m_2) = (m_2^2 - m_1^2)(a^2 + M^2 - m_1^2 - m_2^2)/m_1^2 > 0$. Likewise, $(m_2/m_3)^2 f(m_2) - f(m_3) = (m_3^2 - m_2^2)(a^2 + M^2 - m_2^2 - m_3^2)/m_3^2 > 0$. Taking positive square roots and imposing the trace condition gives $m_2/m_1 < 1 + m_2/m_3 < 2$.

Consequently this canonical universal pure-adjoint candidate cannot reproduce strongly separated first and second quark masses. Degenerate or zero source eigenvalues are boundary cases; they do not evade the nondegenerate hierarchy obstruction. Singlet shifts, nonuniversal heavy or bare Higgs matrices, or nontrivial quark metrics change the assumptions. They are possible mechanisms requiring their own justified contractions, rather than exceptions to this theorem. The symbolic factorization is executable and is checked independently against 80 finite singular-value examples and a traceful counterexample.

All specified one-loop gauge, Yukawa and Higgs evolution

The minimal unbroken action has 24 real scalars: 20 real source fields and four Higgs components. Its colored Yukawa tensors use 24 Weyl flavor/weak indices per color: the three SM families and three vectorlike up and down pairs. Each charge sector has four real coefficients ($y_f, g_{\eta,f}, g_{s,f}, a_f$). Lepton Yukawas are set to zero in this benchmark; the usual SM lepton representations remain in the gauge coefficients. The soft Gaussian fields have no new fermion vertices. This is the renormalizable class fixed by the previous representation audit, rather than the full 251-coefficient mediator scalar action.

An independent full Weyl-tensor calculation and an analytic eight-coefficient expression agree. Define $b_f = g_{\eta,f}^2 + g_{s,f}^2$, $k_f = b_f + 8a_f^2/3$ and $T = 9(y_u^2 + y_d^2)$. The factor nine in T counts three universal families and three colors; this is not a single-top-family approximation. The one-loop coefficients, again suppressing $16\pi^2$, are $\beta(y_f)/y_f = 3(y_f^2 - y_{\text{other}}^2)/2 + T + k_f/2 - 8g_3^2 - 9g_2^2/4 - h_f g_Y^2$, where $h_u = 17/12$ and $h_d = 5/12$.

For either singlet coefficient, $\beta(g)/g = 21b_f + 8a_f^2 + y_f^2 - 8g_3^2 - r_f g_Y^2$; $\beta(a_f)/a_f = 3b_f + 8a_f^2 + y_f^2 - 8g_3^2 - r_f g_Y^2$, with $r_u = 8/3$ and $r_d = 2/3$. The singlet ratio $g_{\eta,f}/g_{s,f}$ is therefore constant at this one-loop order within this action. It remains an input ratio and is not the golden CKM relation. The universal heavy mass obeys $\beta(M_f)/M_f = k_f/2 - 8g_3^2 - r_f g_Y^2$.

The source singlet anomalous dimension is the full two-by-two matrix $18g_i g_j$; retaining its off-diagonal entries matters. Each adjoint block is $6a_f^2 I_8$. The Higgs block is $(T - 9g_2^2/4 - 3g_Y^2/4) I_4$. Quartic evolution combines the exact scalar algebra, these complete wavefunctions, the colored fermion-box polynomial $-6 \sum_f \text{Tr}[(C_f^2 + y_f^2 H^\dagger H) I]$, and the pure-gauge Higgs coefficient $9g_2^2$ to the fourth/ $8 + 3g_2^2 g_Y^2/4 + 3g_Y^2$ to the fourth/ 8 . Source and Higgs mass evolution and the separately running vacuum energy satisfy a direct one-loop Callan–Symanzik identity at independent field points; relative residuals are below 2.1 times 10^{-15} at the retained benchmark.

There are 61 quartics, nine quadratic coefficients, eight Yukawa coefficients, two universal heavy masses and three gauge couplings: 83 nonconstant coefficients. Adding three mediator

masses and six dimension-one currents gives 92. The Gaussian hard-zero restriction is preserved by the stated one-loop superrenormalizable power counting; finite threshold interactions are nevertheless generated. This count is not the complete generic mediator-potential count. Gauge beta coefficients above the six vectorlike thresholds are $(-3, -19/6, 27/2)$, in the order SU(3), SU(2), standard-normalized hypercharge. Below them the SM values are $(-7, -19/6, 41/6)$. The hypercharge convention is not the grand-unified normalization.

The benchmark uses gauge inputs $(1, 0.65, 0.36)$, universal Higgs coefficients $(1.2, 0.57)$, singlet coefficients $(0.1, 0.7)$ and adjoint coefficients 0.8 in each sector. These are declared model inputs, not a measured flavor fit. The integrated 92-coefficient system records tree matching at each scale and stops at the first Higgs quartic zero, $\log(\mu/\mu_0) = 0.5486383814$. The initial $16\pi^2$ beta(λ_H) is -30.85623209 . This is a calculated limitation of these inputs; changing free boundary coefficients can change it. It does not establish a physical CKM renormalization trajectory or a threshold-matched experimental exclusion.

Matching and canonical normalization

The Gaussian heavy-scalar threshold has an exact two-by-two spectral expansion that fixes the leading matrix formulas. At $\mu = M$, with massless light sources and a large-heavy-mass expansion, the threshold coefficients are $16\pi^2 \Delta V_2 = -\text{Tr}(D D^{\text{transpose}})/2$ and $16\pi^2 \Delta V_4 = -\text{Tr}(D H_{\text{EFT}} D^{\text{transpose}})/(2 M^2)$, where D is the heavy-light Hessian block of the source current and H_{EFT} is the light quartic Hessian. A direct spectral calculation checks the field-order expansion independently.

The same soft vertices give an adjoint wavefunction threshold $16\pi^2 \Delta Z_f = \sum_R \dim(R) g_{fR}^2/(4 M_R^2) I_8$. Canonical normalization adds $-(\Delta Z_x) \cdot \text{grad}(V_4)/2$ to the quartic threshold. At the retained finite-mediator inputs the two wavefunction coefficients before division by $16\pi^2$ are 0.1827767223 and 0.1652805891. The complete canonical quartic threshold is projected onto the source basis and checked at independent points; its absolute fit residual is approximately 1.1 times 10^{-12} . Finite light-source masses, higher powers of fields/ M and heavy-quark finite matching are outside this approximation.

The six vectorlike quarks also produce one-loop gauge decoupling, $\Delta(1/g^2) = -\sum_i \Delta b_i \log(m_i/\mu)/(8\pi^2)$. Each heavy up or down family contributes $2/3$ to the QCD coefficient; hypercharge contributions are $16/9$ and $4/9$ respectively. There is no SU(2) contribution because these vectorlike quarks are weak singlets. Matching-scale equality gives zero logarithmic threshold, and the signs and coefficients are tested independently. This matching changes a boundary value; it cannot remove the independent gauge kinetic coefficient. No Sommerfeld constant is derived.

A dimension-eight CP-quality operator

The real adjoint contains a representation-theoretic CP-odd quartic that was excluded from the CP-even scalar potential but must be considered inside Yukawa interactions. Let w_f denote the symmetric square of A_f , and form $O_4 = w_u^{\text{transpose}} (P_5 - P_5^{\text{prime}}) w_d$. Exact $Q(\sqrt{5})$ arithmetic proves family covariance, odd parity under the same generalized CP operation, and $(P_5 - P_5^{\text{prime}})^2 = P_5 + P_5^{\text{prime}}$ in the kinetic metric. The single-sector polynomial vanishes identically; the mixed polynomial does not. It is neutral under both ordinary sector sign charges because each source appears twice.

With a real coefficient c , the interaction $i c O_4 \bar{(Q_L)} H d_R / \Lambda$ to the fourth + Hermitian conjugate is allowed by the specified family and CP symmetries. The coefficient function satisfies $(i O_4)(\text{CP fields}) = (i O_4(\text{fields}))^*$; therefore the full Yukawa operator is CP even despite O_4 being CP odd. Its canonical dimension is eight. On the retained vacuum $O_4 = 0.003462945590$ in declared source units. The universal down Higgs coefficient becomes $y_d + i c O_4 / \Lambda$ to the fourth. The triangular determinant gives the exact tree mass phase $\Delta \theta = 3 \arctan[c O_4 / (y_d \Lambda \text{ to the fourth})]$.

For $c = 1$ and $y_d = 0.57$, the phases at $\Lambda = 1, 10, 100, 1000$ are approximately 0.0182258 , 1.82260×10^{-6} , 1.82260×10^{-10} and 1.82260×10^{-14} radians. Requiring this single contribution to be below a specified reference tolerance of 10^{-10} gives $\Lambda > 116.1911$ in these model units. That tolerance is an illustrative calculation parameter. The cutoff, its relation to physical energy units and the coefficient are not derived. This is one explicit quality operator on the current nonet content, not the earlier complex-link determinant operator or a census of all higher interactions. The renormalizable first-order protection remains valid within its scope.

Higher kinetic interactions retain four physical directions

Both $R = A_u^2$ and $T = A_d^2$ are sign-neutral and CP-compatible source covariants. The existing nine-word Hermitian basis built from R, T has degrees $0, 2, 4, 2, 4, 6, 4, 6, 8$. In orthonormal Hermitian coordinates its determinant is $8 \chi^3$, where $\chi = \text{Im Tr}(R^2 T^2 R T)$. At the retained nonet vacuum $\chi = 1.904868255 \times 10^{-8}$, so these words span all Hermitian quark metrics. Arbitrary infinitesimal Hermitian metric directions can therefore be realized by justified polynomial covariants with real coefficients at this fixed background.

The physical test uses the exact finite six-by-six fermion matrix, canonicalizes a positive down-right metric and computes the light charged-current overlaps by singular vectors. Six Hermitian metric variations are compensated by three diagonal terms so all three down light singular values remain fixed; the up sector and scalar vacuum remain unchanged. The response of $(|V_{us}|^2, |V_{cb}|^2, |V_{ub}|^2, \text{Im}[V_{us} V_{cb} V_{ub}^* V_{cs}^*])$ has four nonzero row-normalized singular values, approximately $(1.67360, 0.99510, 0.41462, 0.19219)$. Two-sided finite variations preserve all six light masses with errors below 10^{-12} and positive metrics above 0.99 .

The heavy masses are allowed to change and their changes are recorded. The earlier six-current test fixed complete light and heavy spectra; this higher-kinetic test makes the narrower light-mass statement. Positive completion by $(I + \sum_i c_i Q_i)^2 + \epsilon I$, $\epsilon > 0$, has scalar degree at most sixteen. A positive metric does not change the tree determinant phase. These higher kinetic vertices are not included in the renormalizable first-order loop theorem. A predictive mechanism must constrain them or supply a justified expansion that suppresses them.

Resolved vacuum selection and quantum feedback

Twenty-four renewed all-source minimizations and six minimizations on the generalized-CP fixed subspace are performed with the Higgs temporarily fixed. CP-fixed candidates are then refined using the full 21-coordinate Higgs/source stationarity equations. The best retained stable CP-conserving candidate has a positive full Hessian minimum above 4 times 10^{-4} , a gradient below 10^{-8} , a CP-fixed residual below 10^{-8} and negligible rephasing-invariant J . Its full-action energy is below the nominated CP-breaking branch by 0.01419555290. The complete coordinates, energies and residuals are stored.

This strict comparison proves that the retained CP-breaking branch is not the global minimum of its declared action. The earlier local-stability result is unaffected. A Gaussian square completion preserves all EFT branch energies at $S = -J/M^2$, so the same comparison applies to that completion at tree level. The absolute global winner over every branch is not proved; an all-field search also finds other lower candidates. A physical symmetry-breaking history cannot be inferred from a local branch alone.

The leading one-loop minimal-action vacuum gradient includes all four Higgs components, the real-source Hessian, twelve Dirac eigenvalues including color, and W/Z terms in Landau gauge at fixed $\overline{MS}$ inputs $\mu = 1$. The linear diagnostic $-H^{-1} \text{grad}(V_{CW})$ has norm 15.5158 times the tree background norm. Its size invalidates a small-displacement treatment for this benchmark. It is not a trustworthy prediction of the new vacuum position. Counterterm choices, changed input coefficients or a new mechanism can modify the result; the present calculation does not silently retune them. The scalar calculation concerns the minimal action and does not include the additional Gaussian mediator modes in the full UV Coleman-Weinberg potential.

Expanding the determinant of the one-loop-corrected fermion matrix gives first-order phases near 10^{-17} and iterated quadratic terms near 10^{-19} radians. These cancellations do not constitute a complete two-loop result: genuine two-loop self energies and finite wavefunction matching are omitted. The receipt labels the determinant expansion explicitly.

The leading high-temperature quadratic potential is calculated from $\text{Tr}(H_{\text{scalar}}) T^2/24$, the colored fermion mass-square trace and the gauge Higgs term. All 24 thermal Hessian eigenvalues are positive at the benchmark, with minimum approximately 0.99998 in the coefficient convention. This supports asymptotic restoration in the leading minimal-action expansion. It is neither a resummed transition calculation nor a choice of CP sign. The extra UV mediators become thermally active above their own thresholds and are not counted in this minimal-action thermal result.

Reproducibility and the next mechanism's obligations

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python python/develop_flavor_frontier.py`. It regenerates three scientific receipts: the exact source proof, full Higgs loop algebra, and `flavor_frontier.json`. The dedicated tests independently verify exact identities and reject a tampered rational coefficient, compare Weyl-tensor and analytic Yukawa evolution, compare fermion boxes to mass-tensor traces, check Callan-Symanzik normalization, re-evaluate finite mass bounds, construct metric responses, and re-evaluate the lower CP-conserving stationary point in the full action. Validation records actual test counts and fresh replay hashes separately; receipt generation is deterministic under the declared numerical environment.

The substantive mechanism-selection conclusions are now precise. Eliminating source singlets alone is excluded by the finite-mass hierarchy theorem. Selecting only one coordinate-supported orientation channel is incompatible with exact scalar-loop closure. Continuous SU(3) unification remains a real relation but lacks the required isolated interior frame. The renormalizable phase theorem requires a separate higher-operator quality mechanism. The retained tree benchmark requires a different vacuum-selection or input structure, and its perturbative vacuum expansion is uncontrolled at $\mu = 1$. An eventual interaction must correlate source labels, quark and mediator vertices, canonical metrics and thresholds while escaping these explicit tests. No target angle or gauge constant is promoted to a prediction by algebraic closure alone.

The primary general-RGE reference is Schienbein, Staub, Steudtner and Svirina, Revisiting RGEs for general gauge theories, arXiv:1809.06797, <https://arxiv.org/pdf/1809.06797>. Its treatment of non-diagonal scalar wavefunctions is relevant to this calculation. For context on Nelson–Barr quality and vacuum questions, see Murai and Nakayama, Revisiting the Minimal Nelson–Barr Model, arXiv:2407.16202v2, <https://arxiv.org/html/2407.16202v2>. The exact projector, restricted-census, pure-adjoint mass-bound and numerical nonet tests here are repository calculations; literature context is not a claim of independent novelty review.